

Bilingual RAG Chatbot for Driving School Client

Project Report — Prepared by Zonia Amer | n8n, Gemini, Qdrant | July 2026

Overview

This project delivers a bilingual (German/English) AI chatbot for a driving school client, capable of answering pricing, timing, and documentation questions accurately across 44 distinct license class pricing structures. The system uses a Retrieval-Augmented Generation (RAG) architecture combined with an agent-based orchestration pattern, built entirely on free-tier infrastructure.

Core Requirement: Zero Hallucination on Pricing

Standard RAG (semantic search + AI summarization) carries real risk when applied to exact numerical data, since embedding-based retrieval can surface approximately-relevant content that a model then paraphrases imprecisely. To eliminate this risk for pricing specifically, all 44 official pricing documents were parsed into structured lookup tables (per-class totals and itemized line items) rather than embedded as free text. Pricing answers are retrieved via exact spreadsheet lookup, not semantic inference — removing hallucination risk entirely for the client's most sensitive data category.

Architecture

- **Intent & Routing:** A central AI Agent reasons over the user's query and decides which specialized tool(s) to invoke, rather than a fixed rules engine.
- **Structured Tools:** Pricing Lookup and Pricing Details (Google Sheets) for exact numeric data; Class Variant Check to detect and resolve ambiguous license class references (e.g. "Class B" → B197 / BE / B96 / etc.).
- **Semantic Tool:** Qdrant vector store, populated via a one-time preprocessing workflow, for unstructured knowledge — opening hours, required documents, company FAQs.
- **Parallel Execution:** The agent can invoke multiple tools within a single turn for multi-part questions.
- **Short-Term Memory:** Session-scoped conversation memory enables multi-turn clarification (e.g. asking which class variant, then resolving it in the next message).
- **Bilingual Handling:** Language is detected per message and responses are generated in the same language, handled natively by the agent's system instructions.

Data Preprocessing

The 44 pricing PDFs were programmatically parsed to extract per-class totals and itemized cost line items (base fees, per-lesson pricing, exam fees, materials) into two structured tables. Four distinct document formatting variants were identified and handled during extraction to ensure complete, accurate coverage across all class types, including flat-rate professional training packages. General knowledge documents (company information, policies, FAQs) were chunked and embedded into a vector store for semantic retrieval.

Technology Stack (Free Tier)

Component	Tool	Purpose
Orchestration	n8n Cloud	Workflow automation and agent orchestration
Language Model	Gemini 2.5 Flash / Flash-Lite	Reasoning, generation, embeddings

Vector Store	Qdrant Cloud (Free Cluster)	Semantic search over unstructured content
Structured Data	Google Sheets	Exact pricing lookup and class variant mapping
File Storage	Google Drive	Source documents and downloadable pricing PDFs

Outcome

The system reliably distinguishes between questions requiring exact data (pricing) and questions requiring contextual understanding (hours, documents, general info), routes accordingly, asks clarifying questions when a request is genuinely ambiguous, and responds in the user's language — all while operating within free-tier API and infrastructure limits.